

Assignment Day – 2

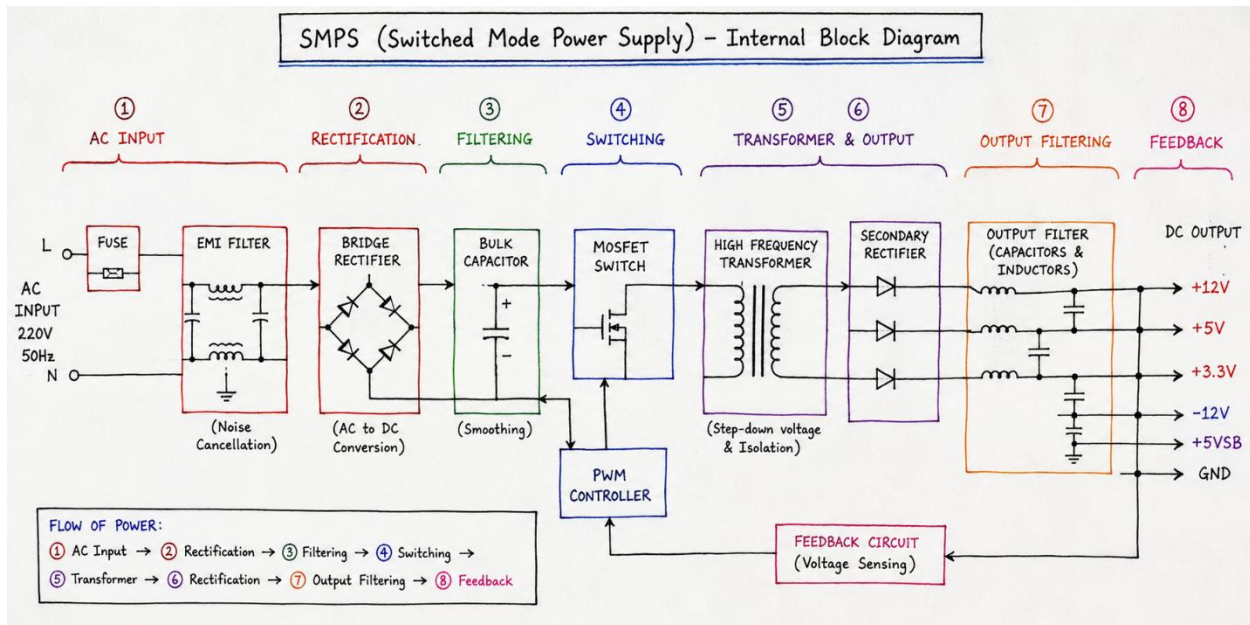
1. List 3 real-world devices in your home or college that use **SMPS (Switched Mode Power Supply)** and 2 devices that use **UPS (Uninterruptible Power Supply)**, explaining in one line why each device needs that type of power supply.



Device	Type	Why it uses this power supply
Desktop Computer	SMPS	Converts AC power into stable DC power for internal components.
LED Monitor	SMPS	Provides the correct DC voltage needed to operate the display.
Wi-Fi Router	SMPS	Converts mains electricity into low-voltage DC for the router.
Desktop Computer (with UPS)	UPS	Keeps the computer running during a power cut and prevents data loss.
Wi-Fi Router (with UPS)	UPS	Maintains internet connectivity when the main power goes off.

2. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.

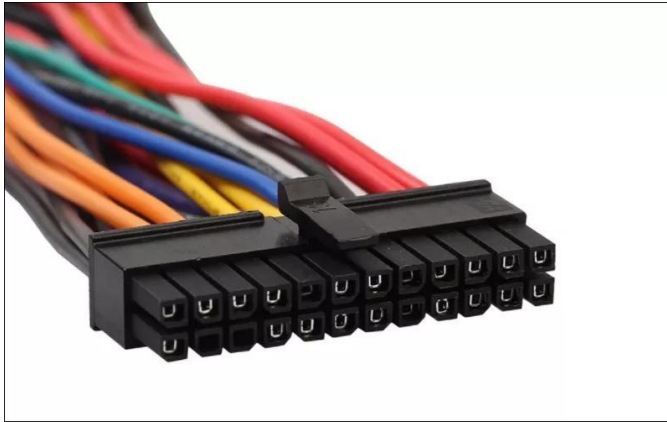
Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.



3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.



a. ATX 24 pin connector



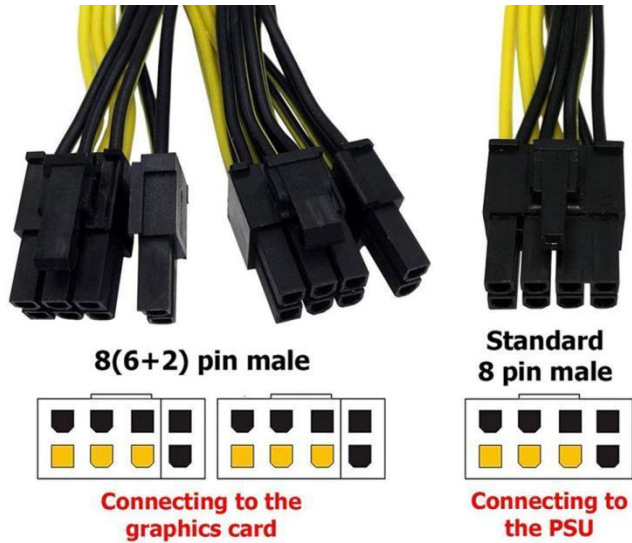
b. SATA Power connector



c. Molex connector



d. PCIe 6/8 Pin



Power Connector	Device it powers	Safety Tip
ATX 24-pin	Powers the motherboard.	Turn off and unplug the PC before connecting or removing it.
SATA Power	Powers HDDs, SSDs, and SATA drives.	Do not force the connector; align it correctly before plugging in.
Molex	Powers older HDDs, fans, and some accessories.	Hold the connector firmly and pull it straight out without pulling the wires.
PCIe 6/8-pin	Provides extra power to a graphics card (GPU).	Make sure the correct PCIe cable is connected before turning on the PC.

4. **Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.**



Steps to Connect SMPS in a Gaming PC

1. **Turn off the PC and unplug it from the power socket for safety.**
2. **Connect the motherboard:**
 - **Use the 24-pin ATX cable from the SMPS.**
 - **Connect it to the 24-pin power socket on the motherboard.**
3. **Connect the CPU:**
 - **Use the 4+4-pin CPU/EPS cable from the SMPS.**
 - **Connect it to the CPU power socket near the processor.**
4. **Connect the graphics card (GPU):**
 - **Use the PCIe 6-pin or 8-pin cable from the SMPS.**
 - **Connect it to the power socket on the graphics card.**
5. **Connect the storage device:**
 - **For a SATA HDD/SSD, use a SATA power cable from the SMPS.**
 - **For an M.2 NVMe SSD, no SMPS power cable is needed because it gets power directly from the motherboard.**
6. **Check all connections:**
 - **Make sure every connector is properly fitted and not loose.**
7. **Connect the SMPS to the wall socket and turn on the PC to check that everything works.**